Do Derivatives Matter?: Evidence From A Policy Experiment*

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We study the impact of derivatives on stock characteristics such as valuation, price efficiency, and liquidity. We resolve the endogeneity issue faced in the extant literature by using an order issued by the Indian market regulator that resulted in delisting of 51 stocks from the derivative segment. Using this policy experiment, we examine the conflicting hypothesis regarding the impact of derivatives on stock fundamentals. We find that excluded firms underperform the market by 4.82% during the event window. Tests using a synthetic control sample also yield directionally similar results. Tobin's Q declines by 6.3% for the excluded firms post exclusion from the derivative segment. We identify decline in price efficiency and reduction in liquidity as channels through which the above phenomenon manifests. We rule out regulatory targeting by employing several placebo and other robustness tests. We conclude that derivatives indeed add value by improving price efficiency and liquidity of a stock.

Key Words: Options market; Tobin's q; Trading Volume.
JEL Classification: G12; G14.

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I Introduction

The question- how derivatives impact a stock's fundamental characteristics such as valuation, price efficiency, liquidity and volatility- has been a subject of academic inquiry for over four decades(Pearson, Poteshman, and White (2007)). Black and Scholes (1973) finding that derivatives are redundant securities whose payoffs can be mimicked by a replicating portfolio, sparked intense research activity on the above question. A number of studies have found that derivatives have a positive impact on the underlying stock by completing markets (Ross (1976)), enhancing informational efficiency of stock prices(Cao (1999); Easley, O'hara, and Srinivas (1998); Roll, Schwartz, and Subrahmanyam (2009); Naiker, Navissi, and Truong (2012); Blanco and Wehrheim (2015)), improving liquidity (Berkman, Eleswarapu, et al. (1998)) and reducing volatility (Damodaran and Lim (1991)). It has also been noted that derivatives alter the distribution of stock returns (Ni, Pearson, and Poteshman (2005)). On the other hand, some studies (Danielsen and Sorescu (2001); Ho and Liu (1997)) find that derivatives depress stock prices as they reduce short sale constraints. Some others have shown that derivatives do not have much of an impact. For example, Muravyev, Pearson, and Broussard (2013) show that option prices do not convey additional information. Vijh (1990) show that large option trades have no impact on stock prices and bid-ask spreads. Given the above conflicting findings Ni, Pearson, and Poteshman (2005) claim that "the literature relating to option introductions and expirations has not shown that equity option trading significantly impacts the prices of underlying stocks."

Such widespread disagreement regarding the impact of derivative securities stem from the lack of clear identification events to empirically examine the impact of derivatives. A major hurdle a researcher in this field faces is the problem of identification. The most commonly used identification strategies are the following;

a. Analysis of stock characteristics such as returns, volatility, liquidity etc before and after derivative listing or delisting (Conrad (1989); Kumar, Sarin, and Shastri (1998); Danielsen and Sorescu (2001)). However, in all these cases, exchanges decide derivative listing and delisting. Mayhew and Mihov (2004, 2005) show that the decision to list a stock in the options segment is endogenously determined by trading volume and volatility over the pre-listing period. Their control group consists of stocks that met the minimum criteria for listing in the options market but were not listed by the exchanges. More importantly, their logit model, which predicts listing based on ex-ante characteristics, also predicts an ex-post increase in volume. This raises the possibility that stocks may be listed in the options market based on their expected volumes or volatility in the future. Danielsen, Van Ness, and Warr (2007) show that derivative listing is accompanied by market-wide changes, which have an impact on stock characteristics. Thus the documented post listing impact of options could be an artifact of selection criteria for listing.

b. Some studies use the extent of derivative trading in a stock to asses the impact of derivatives (Roll, Schwartz, and Subrahmanyam (2009, 2010); Admati and Pfleiderer (1988). The argument here is that mere presence of derivative markets may be insufficient to impact the characteristics of the underlying. A Highly active derivative market is necessary to create any impact. However, as these studies themselves point out, stocks with high options volume may be very different when compared to those with low option volume in many unobservable ways and the option volume may just reflect those differences in characteristics.

We overcome the above identification problem by examining a regulatory order issued by Securities Exchange Board of India[1] due to which 51 stocks were delisted from the derivative segment. On July 23, 2012, SEBI tightened the criteria for continuation of a stock in the derivative segment. Derivative listing criteria in India are based on threshholds, which depend on insider ownership and liquidity in the previous six months. The intent of the regulator was to allow derivative trading in stocks that are widely held and are highly liquid. The thresholds for continuation in the derivative segment were increased by SEBI. We believe that the event was completely devoid of any endogenous factors because of the following reasons;

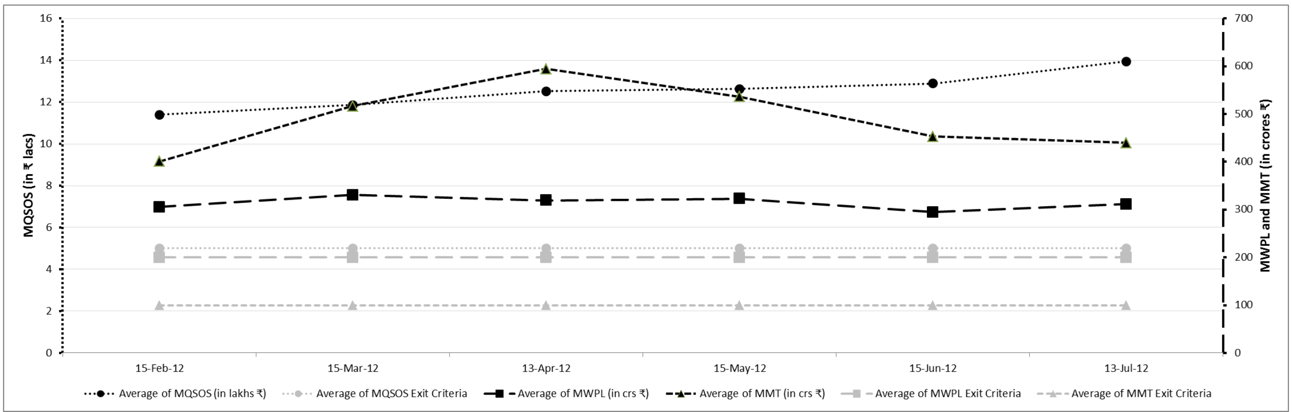
a. First, old as well as new thresholds were based on historical as well as publicly avail-

[1]The Indian market regulator

able information whose impact, if any, is likely to have been priced in (Kaul, Mehrotra, and Morck (2000)).

b. Second, in order to test if there is regulatory targeting as pointed out by (Karlan and Zinman (2009)), we plot the levels of the variables on which the exit decision is based. The insider holding measure is known as Market Wide Position Limit (MWPL) and the liquidity measures are known as Median Quarter Sigma Order Size (MQSOS) and Monthly Market Turnover(MMT).[2] In figure 1, we plot all three variables for excluded stocks for the period of six months before exclusion. As can be seen, MWPL, MMT and MQSOS for excluded stocks are relatively stable in the six months prior to exclusion. Further, there is no dramatic decrease in these measures just prior to SEBI changing the criteria for continued listing. Therefore, it is reasonable to conclude that the exclusion is a result of increase in threshold (grey lines below) rather than any change in excluded stock characteristics.

Figure 1: Trend In Exclusion Criteria



c. Third, it is well known that insider ownership tends to be sticky (Kaserer and Moldenhauer (2008)). From Figure 1, it is clear that MWPL (line with square blocks in between) moves within a very narrow band during the estimation period. Moreover, in the last few weeks before SEBI order, it is almost flat. Given the stability of MWPL, any plausible impact of insider ownership is likely to have been built into prices. Therefore, in order to rule out any residual concerns regarding regulatory targeting, we also study the price reaction of only those stocks that get excluded only because of MWPL deficiency.[3]

[2] The measures have been described in detail in Section V

[3] In other words, when it comes to MQSOS and MMT, these stocks are well above the required

There are 29 such stocks. Because MWPL did not move much during the estimation period, it is difficult to argue that results obtained through this test are influenced by other endogenous factors other than derivative exclusion

d. Fourth, More importantly, unlike the situations considered by the extant literature (Mayhew and Mihov (2004, 2005)), the stock exchanges do not select the stocks to be listed or delisted. Compliance with SEBI regulations is mandatory for all stock exchanges. Therefore, endogenous selection by stock exchanges with an eye on future business opportunities is less of a concern in our setting.[4]

e. Finally, we perform several placebo tests to establish the robustness of our results.

Using the above SEBI order, we first examine the impact of derivatives on stock returns in the short run. From the above discussion, three contradictory hypothesis emerge.

a. First, Conrad (1989) and other show that derivatives add value positively. Under the above hypothesis, an exogenous derivative delisting is likely to lead to negative price reaction.

b. Second, Danielsen and Sorescu (2001) show that derivatives by easing short selling constraints exert downward pressure on stock prices. Under the above hypothesis, derivative delisting is likely to lead to a positive price reaction as short selling constraints increase post delisting.

c. Finally, classical theories (Black and Scholes (1973)) consider derivatives as passive securities, in which case derivative delisting is unlikely to result in any significant price reaction.

We find that the stocks delisted from the derivative segment underperform the broader market by 4.82% during our event window, which spans from 5 days before to 5 days after the event date.[5]. We recognise the fact that the excluded firms may be systematically different from the broader market portfolio. To alleviate the above concern, we create a synthetic control samples by using those stocks that continue to trade in the derivative

threshold

[4]In fact, news paper reports indicate that exchanges opposed such a move

[5]The above result is robust to change in the event window

segment. The weights of the synthetic control sample ensures that our treatment group of delisted firms and the synthetic control groups are comparable on returns. We follow the approach followed in (Acemoglu, Johnson, Kermani, Kwak, and Mitton (2013)) to create synthetic controls. A number of other prominent studies (Abadie and Gardeazabal (2003); Abadie, Diamond, and Hainmueller (2010)) have used the synthetic control approach to arrive at the counter-factual. A control group created this way provides a reasonable estimate of the counter factual situation. Using the synthetic control method, we find that the treatment group firms yield a negative abnormal return of 1.75% during the event window as defined above.[6]. More importantly, stock prices do not recover even after 120 days from the event date. From the above, we conclude that delisting of options leads to loss of shareholder value. These findings are in line with Roll, Schwartz, and Subrahmanyam (2009) and Hu (2014), who show that derivatives indeed add value.

We next investigate the impact of derivatives on long term valuation. Following Roll, Schwartz, and Subrahmanyam (2009), we study the impact of derivative delisting on the Tobin's Q of the sample firms. We estimate a difference-in-difference regression by using stocks who continue to trade in the derivative segment as our control group. We first calculate the difference in change in Tobin's Q between the treatment and the control group firms in the pre regulation change period and then calculate the same difference for the post regulation period. The difference in these two differences provides us with the estimate of the impact of delisting on the treatment firms' Tobin's Q change. We find that the change in Tobin's Q in the post delisting period is lower by 6.3% for the treatment group firms when compared to the control group firms. These results are consistent with our event study results.

Our focus now shifts to the channel of influence. Specifically, we examine two channels:

a. Impact through informational efficiency of the stock price

b. Impact on liquidity

Chen, Goldstein, and Jiang (2007) show that stock prices produce useful information as they aggregate information from various investors, including outside investors. This

[6] Results remain qualitatively similar even when the event window is altered

will help insiders to take effective decisions by learning from the stock prices. Durnev, Morck, and Yeung (2004) measure informational efficiency by looking at the sensitivity of capital expenditure to changes in firms' Tobin's Q. Roll, Schwartz, and Subrahmanyam (2009) follow a similar method to show that derivatives enhance the informational efficiency of a stock by increasing information production. On the other hand, if derivatives attract predominantly noise traders (De Long, Shleifer, Summers, and Waldmann (1990); LONG, Shleifer, Summers, and Waldmann (1990); Shleifer and Vishny (1997)), who engage in positive feedback trading, then such instruments are unlikely to add to price efficiency. Given the nature of the event, our setting allows us cleanly measure the impact of derivatives on informational efficiency of the stock using the method adopted in the above papers. We compare the sensitivity of capital expenditure to change in firm value (as measured by Tobin's Q) before and after delisting for the treatment group and control group.[7] We find a significant decline in the sensitivity of capital expenditure to stock value in the treatment group firms and not among control group firms. This result shows that delisting of derivative contracts also eliminates informed traders, who contribute to price efficiency(Bloomfield, Ohara, and Saar (2009).

To further buttress our findings, we examine the number of analysts following treatment and control group firms before and after delisting. Roll, Schwartz, and Subrahmanyam (2009) show that stocks with higher options activity also attract higher analyst following which in turn increases informational efficiency of a stock. We find that number of analysts following treatment group firms significantly declines after delisting. Results hold even when we implement difference-in-difference specification using firms that continue to be listed as control group. Finally, we also test if earnings surprises increases after derivative listing. If derivatives contribute towards information production, then earnings surprises are expected to increase in the post delisting period. We find that earnings surprises significantly increase for excluded firms in the post delisting period. All the above results show that informational efficiency of the treatment group stock prices declines post delisting from the derivative segments. In other words, derivatives

[7] firms which continue to be in derivative segment

significantly contribute towards information production by inviting trades by informed outside investors.

The second channel that we examine is change in liquidity. Kumar, Sarin, and Shastri (1998) show that options help in enhancing the liquidity of a stock. It is also well known that arbitragers, noise traders and market makers operate both in spot as well as derivatives market to take advantage of any price discrepancies, to hedge positions or for reasons which at best can be termed as noise (Hu (2014). Derivative delisting may drive away many such players who trade in both markets and thereby reduce liquidity in the cash market. We test for change in trading volume in the cash market for affected stocks. We find that the trading volume in the spot market declines significantly post delisting. An overall permanent decrease in the liquidity for the treatment group stocks results in lower valuation (Amihud and Mendelson (1986)). These results indicate that derivatives contribute significantly towards stock liquidity as well.

Finally, we perform a number of tests to establish the robustness of all our results. First, we test for parallel trends (Bertrand, Duflo, and Mullainathan (2002)) and find them in all our difference-in -difference tests. Second, we conduct the following placebo tests to rule out alternative explanations;

a. We conduct all our event studies and regression based tests using several false announcement and implementation dates. In these tests, we keep the exclusion criteria unaltered. We do not find any effects when we conduct tests based on false dates. If our results are driven by the fact that excluded firms had low outsider shareholding or low trading volume pre delisting and not because of delisting, one would expect to get results similar to our main results even in our placebo tests.

b. We also conduct a false limit test where we define our treatment group based on false limits without changing the event date. Here again, we do not find any difference between treatment and control groups.

This paper contributes to the literature on the impact of derivatives by showing that derivatives positively impact stock valuation. The paper also contributes to the literature on informativeness of stock price by showing that price information efficiency is higher in the presence of derivatives than otherwise. Third, unlike existing inferences which are predominantly based on options, our inferences are not just limited to options but can be extended to futures as well. Indian derivative market comprises of both options as well as futures, with both categories trading actively. Finally, and most importantly, this paper exploits a regulatory event that is likely to be devoid of any endogenous information and to that extent produces clean identification, lack of which has been a major challenge for the extant literature in delineating the impact of derivatives.

The rest of the paper proceeds as follows: Section II provides the institutional background.Section III describes the event. Section IV describes the data and summary statistics. Section V develops our main thesis and our empirical strategy Sections VI describes our results. Section VII concludes.

II Background

In this Section, we describe the relevant institutional background.

A Brief History

Stock trading in India has a history of more than 140 years.[8] Bombay Stock Exchange(BSE), which is the oldest stock exchange in Asia, was established in the year 1875. In 1956, the Government of India, after passing the Securities Contract Regulation Act, recognized BSE as a stock exchange. India's first broad-based market index the "Sensex"[9] was introduced in 1986. Another large stock exchange, the National Stock Exchange, was established in 1992 by domestic as well as foreign financial institutions. The NSE soon became the largest stock exchange in India and is currently the 12th largest in the world in terms of market capitalization of listed firms and fourth largest in the world in terms of the annual number of trades in equities.[10] More than 99% of the stock trading in India is effected through these two exchanges.

[8]Source: http://www.bseindia.com/static/about/heritage.aspx?expandable=0

[9]Sensex comprises of 30 large companies in India from different sectors of the economy

[10]NSE's market index is known as CNX Nifty. It represents 50 large companies in India. Source: http://www.nseindia.com/global/content/about$_us/history_milestones.htm$

B Current State

India has highly liquid stock market. Average daily volume of trading in India is to the extent of INR 3 trillion (approximately USD 50 billion.[11] A world bank report[12] shows that between 2010 and 2012, Indian stock turnover to GDP ratio stood at 45.3%, which is higher than many developed markets such as France, Germany and Italy, and almost all emerging markets other than China. The fact that India is now the fastest growing large economy in the world [13] has attracted both domestic as well as foreign institutional capital to Indian markets. Net investment by Foreign Institutional Investors exceeded INR 1 trillion (USD 16.5 billion) in 2014. These facts indicate that India has a well functioning stock market, which has attracted attention from not only domestic investors but also foreign institutional investors.

C Derivatives In India

Equity derivatives were introduced in India in 2000. Unlike the U.S (Mayhew and Mihov (2004)), India started with index futures and options and then introduced stock futures and options in the following year.[14] Very soon, the derivative segment became bigger than the underlying equity segment. As at the beginning of the year 2015, derivative transactions accounted for nearly 91% of all transaction volume in the stock exchanges.[15] One big reason for the relative popularity of derivatives is the lower taxes in that segment. As prominently noted in the literature (Diamond and Verrecchia (1987); Danielsen and Sorescu (2001)), derivatives ease short-sale constraints. Given the short-sale constraints are severe in Indian spot markets (Berkman, Eleswarapu, et al. (1998)), derivatives are relatively more attractive compared to other markets. Roll, Schwartz, and Subrahmanyam (2009) note that in order for the derivatives to have any impact, volume of trading is more important than mere listing of derivatives. Given the high proportion of trading in derivatives, India offers an excellent setting to study the impact of deriva-

[11]Source: www.moneycontrol.com
[12]Source: http://data.worldbank.org/indicator/CM.MKT.TRAD.GD.ZS/countries
[13]Source: IMF, World Bank
[14]Source: www.nseindia.com
[15]Source: www.nseindia.com, www.bseindia.com

tive securities. Another important feature of Indian markets is that both options and futures are actively traded in India. Approximately 30% of derivative trading in India is in options and remaining in futures. This allows us to study the impact of both options as well as futures rather than options alone as is the case with the extant literature.

D Regulation

The Securities and Exchange Board of India (SEBI), was formed on April 12, 1992 [16]with the objective of protecting the interests of investors in securities and to promote the development of, and to regulate the securities market and for matters connected therewith or incidental thereto." All kinds of stock derivative products fall under the regulatory purview of SEBI. SEBI, over the years, has earned a reputation as a responsive and vigilant regulator. The fact institutional failures in India such as broker failures have been few and far between after 2001 shows the efficacy of SEBI's regulations. [17]

III The Event

Given that derivatives are complex instruments and come with high levels of leverage, regulators world over tend to be highly suspicious about them. In the U.S, the SEC took more than seven years to give permanent status to derivative instruments (Mayhew and Mihov (2004)). Initially, the options exchange was given only "experimental status". The SEC declared a "voluntary moratorium" on options trading in 1979 to study the impact of options trading and design appropriate regulatory mechanism. The SEC gave permanent status to options exchanges only in 1980.

SEBI has also been extremely cautious when it comes to derivatives. In fact, a forward trading facility known as the Badla (Berkman, Eleswarapu, et al. (1998)) was blamed by SEBI for "excessive speculation" and volatility in the 90s and was duly banned. Indian policy makers in general are apprehensive of derivative instrument and regularly call for

[16] It was formed in accordance with the provisions of the Securities and Exchange Board of India Act, 1992.

[17] Source: http://articles.economictimes.indiatimes.com/2013-05-24/news/39501911$_{1_i}$*ndian − capital − market − sebi − clearing − corporations*

imposition of restrictions on derivative trading, blaming derivative products for many negative consequences. For example, during periods of inflation, regulators conveniently ban forward trading in agricultural commodities[18] although there is no credible to evidence to suggest that derivative products lead to higher commodity prices (Sahoo and Kumar (2009); Bose (2007)). Despite this regulators in India treat commodity derivatives as whipping boy during the times of inflation (Sahoo and Kumar (2009)). Keeping up with the same spirit and tradition, SEBI, in 2012, came to a conclusion that it is desirable to restrict trading in derivatives in order to protect "market integrity".[19]

On the 22nd of July 2012, SEBI issued guidelines tightening the qualification criteria for continued listing in the derivative segment. One of the criteria was dependent on the amount of outside ownership and other two were liquidity measures. Table 1 provides details about thresholds that prevailed before SEBI notification and the changed level.

Table 1: Old and New Criteria For Continued Listing In The Derivative Segment

We report the old and revised criteria for continued listing in the derivative segment. All figures are reported in INR millions.

Parameter	Earlier	Revised
MWPL	600	2,000
MQSOS	0.2	0.5
MMT	0	1,000

a. Market Wide Position Limit(MWPL): This measure is based on floating stock of a company. The market wide position limit is 20% of the number of shares held by non-promoters in the relevant underlying security.[20] As shown in row 1 of Table 1, the applicable limit of average MWPL over six months was increased from INR. 600 million to INR. 2,000 million.

b. Median Sigma Quarter Order Size (MSQOS): Order size required to move the bid ask mid-point by one fourth of the historical standard deviation of the stock returns. NSE calculates these numbers and puts out the figures in the public domain. As shown

[18] Details about one such instance are given here http://www.legalserviceindia.com/articles/ban$_b$*.htm*

[19] The relevant circular can be found here http://www.sebi.gov.in/cms/sebi$_d$*ata/attachdocs/*1343043461941*.pdf*

[20] Source: http://www.nse-india.com/content/fo/fo$_s$*election.htm*

in row 2 of Table 1, the applicable limit of MSQOS in the last six months for continued listing was revised from INR 0.2 million to INR 0.5 million.

c. Minimum Monthly Turnover (MMT): SEBI introduced a new criterion pertaining to minimum average monthly derivative turnover, which was set at INR. 1,000 million. MMT is calculated based on prior three months turnover.

Stocks that failed to meet any one of the above criteria were excluded from the derivative segment. Stock exchanges were not given any discretion in this regard. However, contracts that existed as on July 22nd 2012 (July 2012, August 2012 and September 2012) continued to trade until expiry. Table 2 (below) reports the number of stocks that get excluded due to various reasons. Please note that 29 stocks got excluded solely because of breaching the extended MWPL threshold, which is dependent on the amount of insider ownership.

Table 2: Number Of Stocks Excluded Due To Various Reasons

We report the number of stocks that get excluded because of failure to meet one or more criteria for continued listing in the derivatives segment.

Reasons	Number of Stocks
Excluded due to MQSOS	5
Excluded due to MWPL	29
Excluded due to MMT	4
Excluded due to MQSOS/MWPL	2
Excluded due to MQSOS/MMT	8
Excluded due to MWPL/MMT	1
Excluded due to MQSOS/MWPL/MMT	1

IV Data and Summary Statistics

In this section, we briefly describe our data sources and relevant summary statistics.

A Data

We obtain most of our data from following four sources;

a. National Stock Exchange(NSE): As noted in Section II, NSE is the largest stock exchange in India. We obtain all price and turnover information from NSE. We cross verify the numbers on a sample basis from multiple other sources. From NSE, we also obtain information pertaining to definition of technical terms such as MWPL (Market Wide Position Limit) and MQSOS (Median Quarter Sigma Order Size), which are typical to Indian markets.

b. Center For Monitoring Indian Economy (CMIE) Prowess: Prowess database, maintained by CMIE, has gained reputation as Indian compustat(Bhue, Prabhala, and Tantri (2015)). The data base provides detailed accounting and financial information for more than 25,000 large and medium companies in India. A number of prominent research articles (Khanna and Palepu (2000); Mehta, Mullainathan, and Bertrand (2002); Gopalan, Nanda, and Seru (2007); Vig (2013)) use the same database. We obtain company-level financial information from Prowess. In particular, we collect data relating to sales, capital expenditure, earnings after interest and tax (EBIT), gross value of assets and cash flows. We note that the information on capital expenditure is available only in annual statements and not in quarterly filings. For the purposes of some tests where a unit of observation is quarter, we calculate capital expenditure by calculating the difference between fixed assets of a quarter and its immediately proceeding quarter.[21]

c. SEBI Web Site: We obtain all relevant SEBI circulars from their web site. From these circulars, we obtain information about derivative listing and delisting norms. We also learn about the effective dates of various regulations from this source.

d. Bloomberg: We obtain data regarding the number of analysts covering each stock from Bloomberg data base. We collect this data at a quarterly frequency.

B Summary Statistics

We report summary statistics in Table 3(below). In the first four columns, we report the number of observations, mean, median and standard deviation of important variables for stocks that were excluded from derivative segment. In the last four columns (column 5 to 8), we report the same for stocks that were retained in the derivative segment. It is clear from the table that retained firms are larger and relatively more liquid when compared

[21]We ignore research and development expenditure as research and development expenditure forms a negligible portion of total expenditure incurred by Indian companies

to excluded firms. For example average market capitalisation (sales) of excluded firms at the time of delisting is INR 4,377 (1,689) whereas the same for retained firms is INR 29,098(5495) million. Similarly stock turnover for the excluded stocks is 79.63 whereas the same for retained stocks is 404.9.

Table 3: SUMMARY STATISTICS

We compare excluded firms and retained firms based on parameters such as sales, market capitalisation, earnings, turnover, market-wide position limit and median quarter sigma order size. We report mean, median, number of observations and standard deviation for all variables. MWPL, MQSOS and turnover are reported for the six month period immediately preceding demisting. Sales and EBIT are for financial year 2011-2012. Price to book ratio is calculated based on the last available financial statement. Reported market capitalisation is as on 21st July 2012. Turnover is calculated by dividing stock volume by floating market capitalisation. All figures in Rupees Million. els.

	Excluded_Firms				Retained_Firms			
Variable	N	Mean	Median	Std Dev	N	Mean	Median	Std Dev
MQSOS (in million INR)	50	13.95	7.63	17.35	154	39.03	23.20	48.24
MWPL (in million INR)	50	311.42	156.05	377.25	154	2,626.14	1,055.89	4,888.85
Mcap (in million INR)	51	43,769.33	21,292.02	60,034.55	155	290,984.50	115,056.73	460,246.82
PB	48	2.79	1.26	4.74	154	3.00	1.82	3.96
Sales(in million INR)	50	16,891.36	7,224.30	33,494.42	154	54,947.54	18,697.60	135,281.41
EBITtoSales (in million INR)	50	0.42	0.22	0.77	154	0.62	0.27	1.54
Turnover in the last six months	51	79.63	50.10	96.73	155	404.89	186.13	708.13

V Empirical Strategy

In this section, we describe our empirical strategy.

A Impact on Valuation

As we note in the introduction, a number of articles (Conrad (1989); Damodaran and Lim (1991); Danielsen and Sorescu (2001) have tested if derivatives improve or worsen stock valuation. We also note that the identification strategy used in the above studies is seriously questioned by subsequent articles (Mayhew and Mihov (2004); Danielsen, Van Ness, and Warr (2007)), which show that the results of earlier studies were influenced by endogenous listing decision of the exchanges. We revisit the question using a exogenous regulatory action as a setting to study the impact of derivatives on stock valuation.

A.1 Market Model

We first estimate a market model to calculate the abnormal returns for the 51 stocks that were excluded from the derivative segment. We use CNX-200 [22] as a proxy for market. We calculate abnormal and cumulative abnormal returns for standard event windows used in the literature. T-statistics computed following the procedure in Boehmer, Masumeci, and Poulsen (1991). T-statistics are adjusted to take into account cross-correlation due to event-date clustering using the methodology described in Kolari and Pynnönen (2010). We estimate the the results using both SEBI announcement day as well as the actual day of exclusion as the event dates.

If derivatives indeed affect value then we expect;

a. negative abnormal returns around the event date;

b. no reversal of returns after the event;

On the other hand if derivatives are redundant, then we do not expect any abnormal returns around the event. Finally, if short selling constraints dominate, then we expect a positive stock price reaction post delisting.

A.2 Synthetic Control Method

In order to address the concern that a broad market index may not represent an appropriate counter factual to the 51 excluded stocks, we develop synthetic controls. We follow the approach adopted by (Acemoglu, Johnson, Kermani, Kwak, and Mitton (2013)) in designing synthetic controls. Our synthetic control sample comes from the pool of 155 retained stocks. The treatment and control groups are matched based on stock returns during the estimation period. Using the method used by (Acemoglu, Johnson, Kermani, Kwak, and Mitton (2013)), we arrive at weights for stocks in the control sample. If results derived using market portfolio as a benchmark are robust, then we expect similar results here as well. In appendix 1, we provide further details regarding construction of synthetic

[22] A broad based index consisting of 200 stocks This is the most liquid Index in India. This represents 50 large stocks from diverse sectors

controls. In appendix 2, we provide details regarding the synthetic control sample as well as the relative weight of each stock in the sample.

A.3 Robustness Tests

As pointed out in the Introduction, we perform a number of robustness tests;

First, as we have pointed out in figure 1, the three exclusion criteria (MWPL, MQSOS and MMT) are based on publicly available historical information. The only new information is the revision in the criteria by SEBI. The revision is completely orthogonal to company fundamentals. As shown by (Kaul, Mehrotra, and Morck (2000)), any plausible price impact of historical information should have already played out. For example, say, TVS Motors has a MQSOS of INR .3 million for the six months ending July 23, 2012 and due to SEBI's notification, the stock gets excluded. Then, it is unlikely that MQSOS, which is public information is likely to move prices after the SEBI announcement. It is difficult to argue that impact of change in MWPL or MOSOS plays out after every six months.

Second, as noted in the introduction, the argument that MWPL, MQSOS or MMT impact is likely to have merit only if there is a decreasing trend in those variables for the affected stocks and not for other stocks. Continuing with the TVS Motors example, say, in the last 10 days MWPL (or MQSOS or MMT) decreases and this decrease brings the average below the new threshold, then one may argue that the price impact post delisting may be an impact of this decrease, which almost coincides or might have influenced SEBI's decision to revise the listing criteria. On the other hand, if the above numbers are steady or increasing or do not show any differential change for the excluded stocks when compared to retained stocks, then it is difficult to make such arguments. In Figure 1, we show that the numbers remain stable during the estimation period.

Third, we perform separate price reaction tests for stocks that get excluded solely because of not meeting the revised MWPL hurdle. It is important to note that MWPL is based on proportion of outside ownership in a stock. This is unlikely to change in a span of six months unless there are some new issues or insider exits.[23] Therefore we expect MWPL to be sticky. Therefore, in order to further rule out regulatory targeting, we estimate the market model separately for stocks that were excluded solely due to lower MWPL. If stocks that get excluded solely due to lower MWPL also react negatively, it is further evidence that the event is indeed endogenous. It is difficult to make a case of regulatory targeting based on MWPL, which is sticky.

Fourth, if the new limit for exclusion is informative, then they are likely to remain so irrespective of SEBI action. We perform placebo tests to test this possibility. We estimate our market model by considering 200 randomly selected placebo dates as event dates and calculate the average CARs around such dates. It is important to note that our treatment group or the placebo excluded sample is likely to be different for different days as we select the sample by applying SEBI's exclusion criteria on each day. For example, on April 1 2011, which is a placebo event day, our sample includes those stocks that that were a part of derivative segment as on April 1 but did not fulfill SEBI's revised criteria for continued listing as per norms declared on July 22nd. Here again, if reaching revised exclusion limits in itself is informative, then one can expect a negative stock price reaction even on the placebo dates. On the other hand, if our results are primarily driven by exclusion of treatment stocks form the derivatives segment, then no price reaction is likely on the placebo dates.

Finally, the above falsification test does not address the concern that only during July 2012, due to some unobservable reasons or reasons known only to SEBI, stocks with lower levels of historical MWPL, MQSOS or MMT, underperform the market. Therefore, the argument could go, that the observed price reaction would have ensued regradless of SEBI action. In order to rule out such a possibility, we perform placebo tests using false limits. We create false limits for MWPL, MQSQS and MMT by keeping the event date unchanged(July 22nd 2012). A new false limit is created at an increment of INR. 10,000 for all three criteria. We estimate the market model by using all possible combination of false limits. The idea here is to test if lower MWPL, MQSOS, or MMT in the previous

[23] We perform a news search and do not find any such events for stocks getting excluded for lack of MWPL

six months indeed translates to lower returns subsequently. If lower levels of MWPL, MQSOS, or MMT lead to lower future returns, then such a phenomenon is likely to manifest even at levels other than SEBI limits. On the other hand, if the results are driven by derivative delisting, then negative returns are likely to manifest only around exclusion limits.

A.4 Tobin Q

It is important to distinguish between short-term price pressure created by the exit of certain type of investors and the long-term impact created by derivatives. Any price impact due to price pressure is likely to reverse itself quickly (Harris and Gurel (1986); Jain, Tantri, and Thirumalai (2014)). On the other hand, if prices move due to the impact of derivatives, then such an impact is likely to be permanent. Following Roll, Schwartz, and Subrahmanyam (2009), we use Tobin's Q as a metric of long-term valuation. Using data spanning from 2 years before to 2 years after delisting, we estimate the following difference-in-difference regression.

$$Y_{\mathrm{ij}} = \alpha + \nu_i + \delta_j + \theta_{sj} + \beta_1 * \mathrm{After} * \mathrm{Treat} + \beta_2 * \mathrm{After} + \beta_3 * \mathrm{Treat} + \beta_4 * X_{ij} + \epsilon_{ijs} \tag{1}$$

Each observation is at a firm-quarter level. The total sample includes firms that are excluded from the derivative segment and those that continue to have derivatives traded on them even after the SEBI order. The dependent variable is the change in Tobin's Q of firm i as at the end of quarter j when compared to the same as at the end of quarter j-1. The independent variable After is a dummy variable that takes a value of 1 if quarter i lies in the post delisting period and zero otherwise. Treat is a dummy that takes a value of 1 if stock i under consideration is an excluded stock and zero otherwise. Xij represents a vector of firm level controls that include controls for size, trading volume and profitability. We also include industry and time fixed effects in order to absorb any time trend as well as time invariant firm-level characteristics.

The main independent variable of interest is the interaction between After and Treat dummies. This represents a difference-in-difference measure and can be represented as follows;

$$\begin{aligned}\beta_1 &= \overline{Y}_{\text{Delisted stocks}} - \overline{Y}_{\text{other stocks}}\Big|_{\text{Post delisting period}} \\ &\quad - \overline{Y}_{\text{Delisted stocks}} - \overline{Y}_{\text{other stocks}}\Big|_{\text{Pre Delisting Period}} \end{aligned} \tag{2}$$

If derivatives add long term value, then we expect the difference-in-difference coefficient to be positive and significant. Otherwise, we expect it to be statistically indistinguishable from zero.

B The Channel

We then move on to examine the channels through which derivatives influence stock valuations. Specifically, we examine if derivatives;

a. enhance price efficiency of stock prices by facilitating informed trading;

b. add to liquidity of stock

B.1 Price Efficiency- Association Between Capex and Stock Returns

To test price efficiency, we follow Roll, Schwartz, and Subrahmanyam (2009). The idea here is that if the stock prices reflect information, which is not available to the management privately, then there is likely to be high association between capital expenditure and signals emanating from stock prices. Tobin's Q is used as a proxy for such a signal. We measure the association between capital expenditure in one period and Tobin's Q in the previous period by estimating the following regression equation separately for the pre and post periods;

$$Capex_{\mathrm{ij}} = \alpha + \nu_i + \delta_j + \theta_{sj} + \beta_1 * \mathrm{Q} * \mathrm{Treat} + \beta_2 * \mathrm{Q} + \beta_3 * \mathrm{Treat} + \beta_4 * X_{ij} + \epsilon_{ijs} \quad (3)$$

Each observation is at firm-year level. The dependent variable is the logarithm of capital expenditure incurred by a firm i during year t. Other terms have the same meaning as defined before. If derivatives enhance price efficiency of the stock, then we expect a lower association between capital expenditure and Tobin's Q in post period when compared to pre period.

B.2 Price Efficiency- Number Of Analysts Following

Following (Damodaran and Lim (1991)), we also test if there is any change in the number of analysts following excluded stocks in the post delisting period when compared to pre delisting period. It has been recognised in the literature that analysts (Asquith, Mikhail, and Au (2005)) contribute towards both production of new information as well as interpretation of existing information. It has also been hypothesised that informed investors prefer derivatives for various reasons (Chakravarty, Gulen, and Mayhew (2004)).Given the above, we test if delisting results in reduced interest for the stock within analyst community. Here again, we estimate a difference-in-difference regression similar to equation (1) above. The dependent variable is the number of analysts covering a stock i during a quarter j. All other terms remain same as in equation (1).

A negative and significant value for the difference-in-difference term would indicate that the analyst interest in the excluded stocks reduces post delisting. Such a result strengthens our view that derivatives lead to increased informed investor participation and hence their exclusion results in loss in informational efficiency. On the other hand, if derivatives have no such impact as (Vijh (1990)) shows, then we expect the difference-in-difference coefficient to be insignificant.

B.3 Price Efficiency- Standardised Unexpected Earnings

Earnings surprises are less likely in informationally efficient stocks (?). We now go on tests if derivative delisting increases earnings surprises. In the absence of data on consensus estimates in India, we use the Standardized Unexpected Earnings(SUE), which is calculated in line with ?. The earnings shock as measured by the Standardized Unexpected Earnings(SUE) is the normalized gap in the announced profit over analysts expectation. We calculate SUE as follows:

$$SUE_{it} = (Profit_{it} - Profit_{i,t-4})/\sigma_i, \tag{4}$$

where, $Profit_{it}$ and $Profit_{i,t-4}$ are the quarterly profits at t^{th} and $(t-4)^{\text{th}}$ quarter respectively for firm i and σ_i equals the standard deviation of profits.

To understand the impact of derivatives on price efficiency, we estimate an equation similar to equation (1) with dependent variable being the SUE for a quarter. If derivatives indeed improve price efficiency, then we expect the SUE to be positive and significant for delisted stocks post derivative delisting.

B.4 Liquidity

Finally, we test for the impact of derivatives on the overall liquidity of stock. It is possible that a number of traders prefer trading in derivatives over spot markets. The reason behind such a preference could be leverage offered by the derivatives markets, which may be hard to replicate through borrowing due to the presence of market frictions. Such traders could be either informed, liquidity or noise traders. If derivative delisting drives away such traders as in Berkman, Eleswarapu, et al. (1998), then derivative delisting is likely to lead to loss of value due to reduced liquidity.

We test for change in liquidity by calculating abnormal volume around the event. We calculate abnormal volume as follows;

a. for any given trading day i, we first calculate the average trading volume over the 60 days immediately preceding day i.

b. The difference between the volume on day i and the average as calculated above represents our measure of abnormal volume.

We estimate the following regression equation;

$$Ab_{V}ol_{\text{ij}} = \alpha + \nu_i + \delta_j + \theta_{sj} + \beta_1 * \text{Treat} + \beta_2 * X_{ij} + \epsilon_{ijs} \tag{5}$$

This is a cross sectional regression with abnormal volume of a stock i during a window j being the dependent variable. The total sample includes retained as well as excluded stocks. Treat is a dummy that takes the value of one for excluded stocks and zero for other stocks. Xij refers to firm level control variables similar to ones used on equation (1).

VI Results

In this section we present our results using empirical strategy described in Section V.

A Market Reaction

The results regarding market reaction to exclusion of stocks from the derivative segment are presented in Graph 2 and Table 4. We report daily averages for 5 days before and 5 days after the event date. We estimate a market model, with CNX-200 as the benchmark.

In the above table, event time is measured as the number of trading days from the event date. N (in column 2) refers to number of treatment stocks. AR stands for abnormal return generated by the market model for a stock i as on day d. CAR is the cumulation of AR from day -5 to day d. Std CS test refers to t-statistics computed following the procedure in Boehmer, Masumeci, and Poulsen (1991). T-statistics are adjusted to take into account cross-correlation due to event-date clustering using the methodology described in Kolari and Pynnönen (2010).

In column 3 (4), we report the mean (median) abnormal returns. In column 4 (5),

Table 4: MARKET REACTION

The table shows the price reaction of sample stocks around the SEBI notification date an event-by-event basis. Event time is measured as the number of trading days from the event date. N (in column 2) refers to number of excluded stocks, for which we have data. AR stands for abnormal return generated by the market model for a stock i as on day d. CAR is the cumulation of AR from day -5 to day d. Std CS test refers to t-statistics computed following the procedure in Boehmer, Masumeci, and Poulsen (1991). T-statistics are adjusted to take into account cross-correlation due to event-date clustering using the methodology described in Kolari and Pynnönen (2010).

Stock Price reaction around event day using MM Model-CNX200

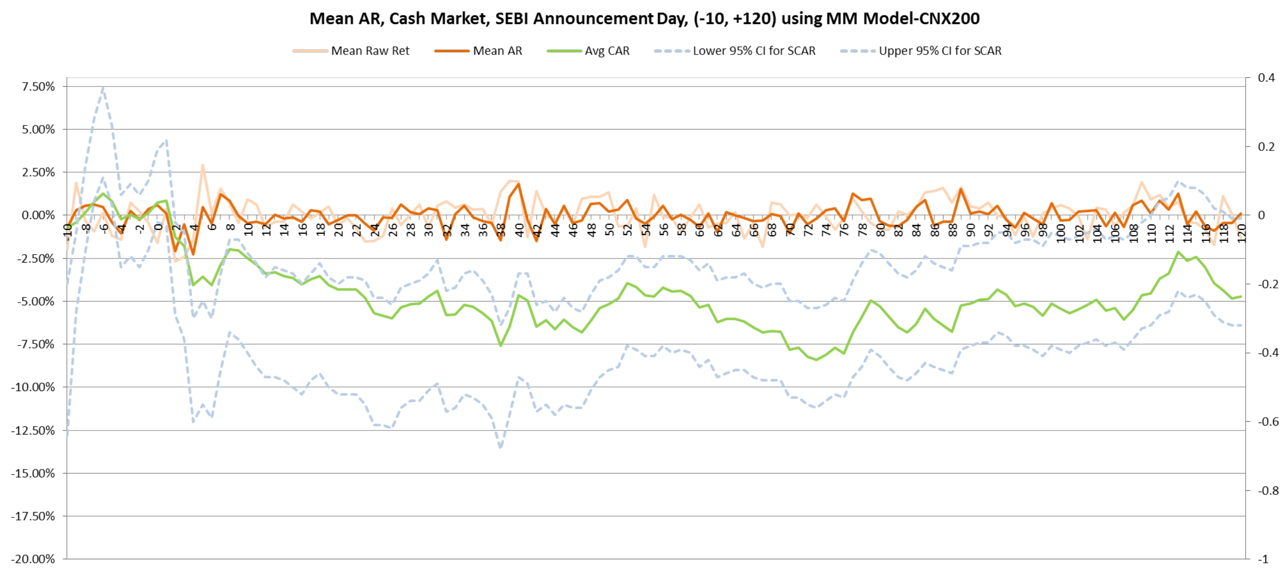
		AR		CAR			
Event Time	**N**	**Mean**	**Median**	**Mean**	**Median**	**CS P-val**	**SR P-val**
-5	50	-0.42%	-0.58%	-0.42%	-0.58%	27.10%	4.51%**
-4	50	-1.04%	-0.97%	-1.46%	-1.53%	0.324%***	0%***
-3	50	0.25%	0.22%	-1.21%	-1.25%	2.77%**	0.002%***
-2	50	-0.27%	-0.55%	-1.49%	-1.76%	1.67%**	0%***
-1	50	0.38%	0.18%	-1.11%	-1.45%	10.30%	0.028%***
0	50	0.59%	0.23%	-0.51%	-1.17%	51.40%	8.85%*
1	50	0.10%	-0.18%	-0.41%	-1.34%	69.10%	23.30%
2	50	-2.09%	-1.94%	-2.51%	-2.63%	5.79%*	0.143%***
3	50	-0.53%	-0.47%	-3.03%	-3.04%	6.02%*	0.003%***
4	50	-2.27%	-2.12%	-5.31%	-4.86%	0.308%***	0%***
5	50	0.49%	0.54%	-4.82%	-4.86%	0.462%***	0%***

we report the mean (median) cumulative abnormal returns. The cumulation starts from day -5. In column 5(6), we report appropriate p values for mean (median) cumulative abnormal returns.

From the table, it is clear that stock reaction around the event is negative. Mean (Median) cumulative abnormal return during -5 to +5 day window that we consider is -4.82% (-4.86%). Both the above numbers are economically as well as statistically significant. We obtain similar results even when we use Nifty, which is the most widely tracked Indian Index comprising of fifty large stocks, as a market benchmark.

We next plot both abnormal and cumulative abnormal returns in figure 2. Here we cover a window spanning from 10 days before and 120 days after the event day. The idea is to see if the negative shock shown in Table 4 gets reversed quickly or does it persist for a longer time. As we have pointed out in Section V, a persistent negative shock strengthens our inference that derivatives indeed add value.

Figure 2: Market Reaction



In the above figure, green line represents cumulative abnormal returns. Notice that this line remains in the negative territory even after 120 days from the event date. Also notice a sharp negative slide after day 46. This is the day when all existing derivative contracts expire. From the above, we infer that negative abnormal returns caused by derivative exclusion do not reverse immediately after the event and hence the reaction that we have documented is unlikely to be an instance of price pressure. We conduct further tests to test if the price reaction is permanent.

B Synthetic Control Model

We now estimate abnormal returns using synthetic controls. The results are reported in Table 4 and graphically depicted in Figure 3. We report daily averages for 5 days before and 5 days after the event date.

Table 5: MARKET REACTION USING SYNTHETIC CONTROL METHOD

The table shows the price reaction of sample stocks around the SEBI notification date an event-by-event basis. Here, we use synthetic control method in order to form a control group. Event time is measured as the number of trading days from the event date. N (in column 2) refers to number of excluded stocks, for which we have data. AR stands for abnormal return generated by the market model for a stock i as on day d. CAR is the cumulation of AR from day -5 to day d. Std CS test refers to t-statistics computed following the procedure in Boehmer, Masumeci, and Poulsen (1991). T-statistics are adjusted to take into account cross-correlation due to event-date clustering using the methodology described in Kolari and Pynnönen (2010).

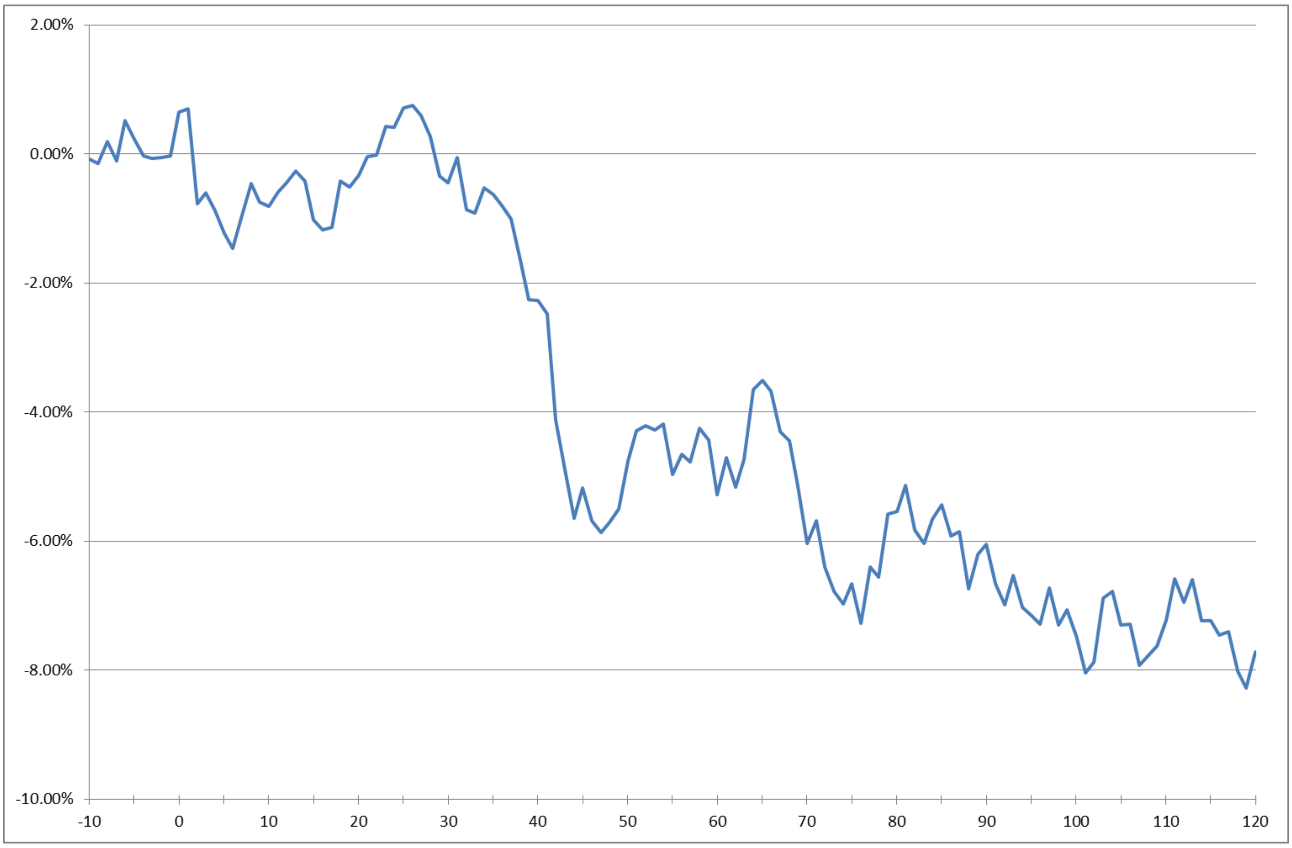
Stock Price reaction around event day using MM Model-Synthetic Control

		AR				CAR			
Time	**N**	**Mean**	**Md**	**CS Pval**	**SR Pval**	**Mean**	**Md**	**CS Pval**	**SR Pval**
Panel A (-5, +5)									
-5	50	-0.28%	-0.42%	12.88%	3.04%**	-0.28%	-0.42%	12.88%	3.04%**
-4	50	-0.26%	-0.22%	20.00%	4.32%**	-0.54%	-0.61%	7.08%*	1.16%**
-3	50	-0.04%	-0.10%	78.10%	92.58%	-0.58%	-0.43%	5.92%*	5.12%*
-2	50	0.00%	-0.29%	99.30%	15.72%	-0.58%	-0.45%	9.88%*	7.24%*
-1	50	0.03%	-0.16%	91.22%	86.98%	-0.55%	-0.57%	12.84%	12.68%
0	50	0.68%	0.47%	0.84%***	21.80%	0.13%	-0.26%	94.74%	37.67%
1	50	0.05%	0.13%	84.86%	33.87%	0.19%	-0.22%	88.70%	62.43%
2	50	-1.47%	-1.36%	0.00%***	0.00%***	-1.29%	-1.51%	3.60%**	3.36%**
3	50	0.16%	0.18%	64.51%	71.15%	-1.13%	-0.95%	9.00%*	18.32%
4	50	-0.27%	-0.39%	29.07%	6.32% *	-1.40%	-0.88%	5.68%*	12.72%
5	50	-0.35%	-0.25%	29.75%	62.99%	-1.75%	-1.43%	2.76%**	3.36%**

In table 5, we report abnormal returns and cumulative abnormal returns using synthetic control method. Event time is measured as the number of trading days from the event date. N (in column 2) refers to number of excluded stocks, for which we have data. AR stands for abnormal return generated by the market model for a stock i as on day d. CAR is the cumulation of AR from day -5 to day d. Std CS test refers to t-statistics computed following the procedure in Boehmer, Masumeci, and Poulsen (1991). T-statistics are adjusted to take into account cross-correlation due to event-date clustering using the methodology described in Kolari and Pynnönen (2010). In line with earlier results, we find that the event leads to significant negative abnormal returns. Stocks that are excluded from the derivative segment yield a average (median) negative cumulative abnormal return of 1.75 % (1.43%) between event date and five days after that. The negative abnormal returns sustain even after 60 days from the event date.

We plot both abnormal and cumulative abnormal returns using synthetic control sample as the benchmark in figure 3. Here we cover a window spanning from 10 days before and 120 days after the event day. The idea is to see if the negative shock shown in Table 5 gets reversed quickly or does it persist for a longer time. As we have pointed out in Section V, a persistent negative shock strengthens our inference that derivatives indeed add value.

Figure 3: Market Reaction Synthetic Controls



In the above figure, the blue line represents cumulative abnormal returns. Notice that this line remains in the negative territory even after 120 days from the event date. Also notice a sharp negative slide after day 48. This is the day when all existing derivative contracts expire. From the above, our inference that the negative abnormal returns caused by derivative exclusion do not reverse immediately after the event and hence the reaction that we have documented is unlikely to be an instance of price pressure gets strengthened.

C Reaction Of Stocks Excluded Solely For Not Fulfilling MWPL Criterion

We show through Figure 1 presented in the Introduction that MWPL, which depends on proportion of insider ownership, remains almost flat during the estimation period. We also discuss in Section V that the possibility of regulatory targeting is lowest for stocks excluded solely for not meeting revised MWPL threshold. We point out in Table 2 that 29 stocks get excluded for not fulfilling minimum MWPL requirement alone. We estimate the market model only for those stocks that get excluded the derivative segment due to breach of minimum MWPL requirement. Results are reported in Tables 6.

From the table, it is clear that stock reaction around the event is negative. Mean (Median) cumulative abnormal return during -5 to +5 day window that we consider is -6.61% (-7.28%). Both the above numbers are economically as well as statistically significant. The above result rules out the possibility of regulatory targeting. We cannot estimate the price reaction separately for stocks that get excluded for reasons other than MWPL as such stocks are very few in number.

In order to examine the long term reaction of stocks that get excluded because of MWPL, we plot the stock returns around event date in figure 4.

The green line represents cumulative abnormal returns. It is clear from the figure that negative abnormal returns tend to persist even after 120 days after the date of exclusion. Excluded stocks yeild a cumulative abnormal negative return of nearly 11% during 120 days after the event day. This rules out price pressure driving our results. Given that

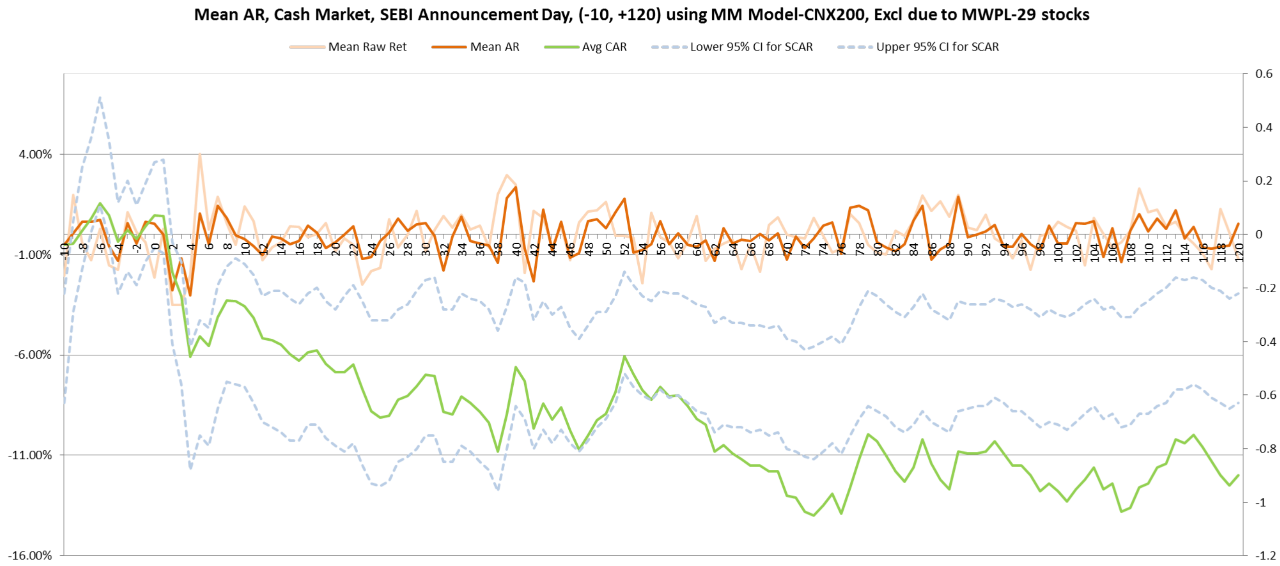
Table 6: Market Reaction

The table shows the price reaction of sample stocks around the SEBI notification date an event-by-event basis. Here we consider only those stocks that get excluded solely because of not meeting revised MWPL criteria. Event time is measured as the number of trading days from the event date. N (in column 2) refers to number of excluded stocks, for which we have data. AR stands for abnormal return generated by the market model for a stock i as on day d. CAR is the cumulation of AR from day -5 to day d. Std CS test refers to t-statistics computed following the procedure in Boehmer, Masumeci, and Poulsen (1991). T-statistics are adjusted to take into account cross-correlation due to event-date clustering using the methodology described in Kolari and Pynnönen (2010).

Stock Price reaction for stocks excluded due to breach of MWPL

		AR		**CAR**			
Event Time	**N**	**Mean**	**Median**	**Mean**	**Median**	**CS P-val**	**SR P-val**
-5	29	-0.59%	-0.88%	-0.59%	-0.88%	20.60%	5.11%*
-4	29	-1.30%	-1.07%	-1.89%	-2.28%	0.39%***	0.001%***
-3	29	0.56%	0.73%	-1.33%	-1.33%	9.19%*	0.491%***
-2	29	-0.45%	-0.67%	-1.78%	-2.19%	4.29%**	0.018%***
-1	29	0.63%	0.43%	-1.14%	-1.46%	24.90%	1.06%**
0	29	0.55%	0.07%	-0.60%	-1.29%	60.40%	21.10%
1	29	-0.02%	-0.19%	-0.62%	-1.71%	66.90%	33.40%
2	29	-2.79%	-3.01%	-3.41%	-3.34%	5.49%*	0.296%***
3	29	-1.19%	-2.66%	-4.60%	-5.20%	4.7%**	0.001%***
4	29	-3.05%	-2.96%	-7.65%	-9.03%	0.209%***	0%***
5	29	1.04%	0.74%	-6.61%	-7.28%	0.526%***	0%***

Figure 4: Market Reaction



there is very little chance of regulatory targeting, this result represents a cleaner measure of impact of derivatives on stock prices.

D Stocks Excluded Because of MWPL- Discontinuity

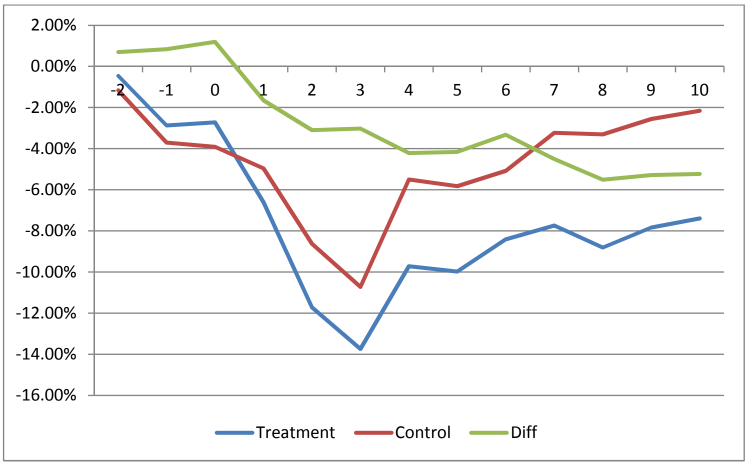
We further tighten the test performed in Section VI.C by designing a test in the spirit of regression discontinuity. We compare stocks that got excluded due to MWPL shortfall with those that just managed to clear the MWPL threshold. Specifically, stocks with MWPL between INR. 200 to 250 million (200 being the cut-off) form our counter-factual. From the excluded group, we exclude stocks with very low MWPL (less than INR. 100 million) in order to maintain comparability. After the above filtering, stocks with MWPL between INR. 100 million become part of our treatment group and stocks having MWPL between INR. 200 and 250 million form our control group. Such tight restriction reduces the number of observations considerable. We are left with only 27 stocks in the treatment group and 10 stocks in the control group. This prevents us from performing parametric statistical tests. Nevertheless, we plot median raw returns for the treatment and control portfolios.

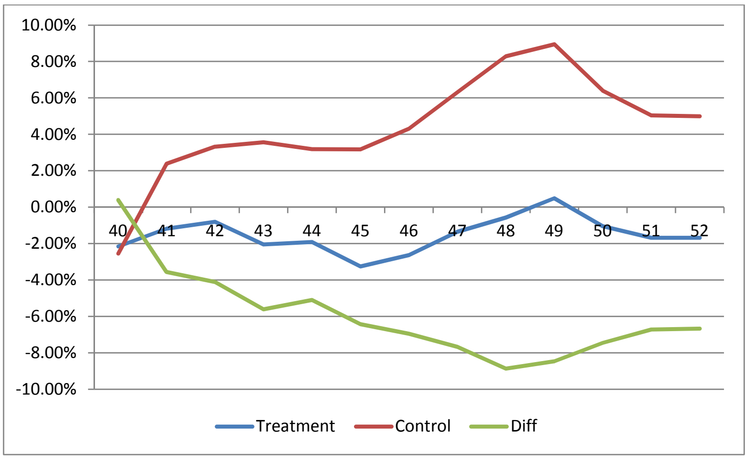
The results are presented in figure 5

The upper panel looks at price reaction of two portfolios and the difference between

Figure 5: Raw Returns For Discontinuity Tests Based on MWPL

In both the figures, horizontal axis denotes days and the vertical axis denotes raw returns. Blue line represents the treatment group, red line represents the control group and green line represents the difference in returns between the two portfolios.





them around the announcement dates. The median return for the treatment portfolio is lower by about 4% when compared to the control portfolio during -1 to +5 window. In the lower panel, we look at reaction at the time of actual exclusion. We find similar results as above. Although the identification is very clean here, we use this as a robustness test only owing to low number of observations.

E Placebo tests

Here we present the results of two placebo test described in Section V.

E.1 False Dates

We estimate the market model for placebo event dates by following the procedure described in Section IV. We calculate the average Abnormal Return and Cumulative Abnormal Return from all placebo event dates. For the sake of brevity, we do not report the above results.

On an average, we do not find any significant stock movement around the placebo dates. We find that the average of all placebo tests indicate that, in general, having either MWPL or MQSOS within the range that lead to exclusion of 51 stocks in July 2012, does not convey any price sensitive information. If it did, negative reaction would have manifested even in the placebo tests.

E.2 False Limits

We perform the false limit test described in Section 5. As before, we calculate the average Abnormal Return and Cumulative Abnormal Return. For the sake of brevity, we do not report the above results.

In line with our main hypothesis that the results reported in this paper are driven by derivative exclusion and not by either regulatory targeting or due to lower trading volumes, we find no significant reaction around the event days when we use false limits. Therefore it cannot be said that our 51 excluded stocks lost value due to them having lower MWPL or MQSOS.

Finally, as we explain in the Introduction and depict in figure 1, there does not seem to be a falling trend of MWPL or MQSOS for the delisted stocks.

F Impact on Long Term valuation

In order to assess the impact of derivatives on long term valuation, we assess the impact of derivatives on permanent valuation of a company. Following Roll, Schwartz, and Subrahmanyam (2009), we use Tobin's Q as a metric of valuation. We estimate equation (1) as described in section V. The results are reported in Table 6.

Table 7: Change in Tobin's Q before and After the Event

The table shows the impact of derivative exclusion on long term value of a stock. The total sample includes firms that are excluded from the derivative segment and those that continue to have derivatives traded on them even after the SEBI order. The dependent variable is the Change in Tobin's Q of firm i as at end of quarter j when compared to the same as at the end of quarter j-1. The independent variable after is a dummy variable that takes the value of 1 if quarter i lies in the post delisting period and zero otherwise. Treat is a dummy that takes the value of 1 if the stock i under consideration is an excluded stock and zero otherwise. The main independent variable of interest is the interaction between After and Treat dummies. We also include firm level controls for size(market cap), insider holding(MWPL) and liquidity (MQSOS). In column 1 we do no include any fixed effects. In column 2, we include firm fixed effects and in column 3, we include both firm and time fixed effects. T-statistics are reported in parenthesis. ***, **, * represents statistical significance at the 1%, 5% and 10% levels.

VARIABLES	(1) Chg_Tobinsq	(2) Chg_Tobinsq	(3) Chg_Tobinsq
Treatment	-0.001		
	(0.019)		
Post_event	0.081***	0.077***	-0.753***
	(0.013)	(0.016)	(0.071)
Int_treat_post	-0.039*	-0.063**	-0.063**
	(0.021)	(0.029)	(0.028)
Observations	2,956	2,956	2,956
R-squared	0.007	0.013	0.282
Adj R-squared	0.00507	0.0127	0.282
Observations _id	204	204	204
Control Vars	YES	YES	YES
Time FE	NO	NO	YES
Firm FE	NO	YES	YES

The table shows the impact of derivative exclusion on long term value of a stock. The total sample includes firms that are excluded from the derivative segment and those that continue to have derivatives traded on them even after the SEBI order. The dependent

variable is the Change in Tobin's Q of firm i as at end of quarter j when compared to the same as at the end of quarter j-1. The independent variable after is a dummy variable that takes the value of 1 if quarter i lies in the post delisting period and zero otherwise. Treat is a dummy that takes the value of 1 if the stock i under consideration is an excluded stock and zero otherwise. The main independent variable of interest is the interaction between After and Treat dummies. We also include firm level controls for size(market cap), insider holding(MWPL) and liquidity (MQSOS). In column 1 we do no include any fixed effects. In column 2, we include firm fixed effects and in column 3, we include both firm and time fixed effects.

As noted in Section V, we are primarily interested in the interaction between Tobin's Q and the post dummy. As can be seem in both column 1 and 2, the interaction term shows that Tobin's Q is lower for treatment firms in the post exclusion period. Tobin's Q falls by a statistically as well as economically significant 6.3% for excluded stocks in the post exclusion period. The above result indicates that the impact of derivative exclusion is likely to have a permanent impact on valuation of a company.

G Channels of Influence

In Section V, we hypothesize that derivatives influence stock valuations by altering price efficiency and by influencing liquidity.

G.1 Price Efficiency Test Using Capital Expenditure

We first estimate equation (3) in the manner specified in Section V. The idea here is to check if the ability of stock prices to influence real decisions gets altered by the existence of derivatives. The results are presented in Table 7.

In column 1 and 3 and 5, we estimate equation (2) for pre delisting period. In columns (2), (4) and 6, we do the same for post delisting period. We look for association between capital expenditure in period t and Tobin's Q in period $t-1$. We find that the association between capital expenditure and Tobin's Q is positive although statistically insignificant in the pre delisting period. In the post delisting period on the other hand the associ-

Table 8: Impact Of Derivatives On Price Efficiency

The table shows the impact of derivative on price efficiency. Each observation is at firm-year level. The dependent variable is the logarithm of capital expenditure incurred by a firm i during year t. All other terms have the same meaning as before. We estimate the following regression equation separately for the pre and the post periods;

$$Capex_{\mathrm{ij}} = \alpha + \nu_i + \delta_j + \theta_{sj} + \beta_1 * \mathrm{Q} * \mathrm{Treat} + \beta_2 * \mathrm{Q} + \beta_3 * \mathrm{Treat} + \beta_4 * X_{ij} + \epsilon_{ijs} \tag{6}$$

In columns 1, 3 and 5 we report results for pre period and in columns 2, 4 and 6, we report the same for the post delisting period. In columns 1 and 2, we do not include any control variables. In columns 3 and 4, we include firm level control varaibles. In columns 5 and 6, we include both firm as well time fixed effects along with firm level control variables.

VARIABLES	(1) ln_capex	(2) ln_capex	(3) ln_capex	(4) ln_capex	(5) ln_capex	(6) ln_capex
Treatment	-0.791	-0.633*	-0.080	0.027		
	(0.576)	(0.343)	(0.584)	(0.376)		
Tobinsq	-0.177**	0.023	-0.291**	-0.154	-0.007	-0.237
	(0.077)	(0.124)	(0.113)	(0.118)	(0.091)	(0.175)
Lagtobinsq	0.086	-0.068	0.117	-0.022	0.357**	0.049
	(0.073)	(0.111)	(0.099)	(0.114)	(0.140)	(0.158)
Int_lagq_treat	-0.014	-0.282**	-0.085	-0.286**	0.060	-1.213*
	(0.308)	(0.136)	(0.313)	(0.126)	(0.319)	(0.662)
Observations	298	276	297	273	297	273
R-squared					0.109	0.151
Observations	182	164	182	161	182	161
Adj R-squared	0.0168	0.0293	0.0266	0.0351	0.109	0.151
Control Vars	N0	N0	YES	YES	YES	YES
Time FE	N0	N0	N0	N0	YES	YES
Firm FE	N0	N0	N0	N0	YES	YES

ation between the two variables turns negative and both statistically and economically significant. The results hold even after controlling for time invariant firm level factors and also time trend. This shows that signals emanating from the stock price become less informative for corporate managers in their capital expenditure decisions in the post delisting period when compared to post delisting period.

G.2 Price Efficiency Using Analyst Coverage

If a significant chunk of informed investors use only the derivatives route for exploiting the information at hand, then derivative exclusion is likely to see exit of such investors from excluded stocks. Consequently, analyst coverage on excluded stocks is likely to decrease. We test the above phenomenon by estimating regression equation (1) with number of analysts following a stock in a quarter as a dependent variable. The results of the above regression are reported in Table 8.

Each observation is at a firm quarter level. Our focus here is on the difference in difference term, which represents the change in difference between analyst coverage between excluded and retained firms in the post period when compared to pre period. We find that on average nearly four (column1 of table 8) analysts stop covering the excluded firms because of derivative exclusion. The number drops to nearly 3 (columns 2-4), when we include other firm level controls and fixed effects. Given that the average number of analysts covering a stock is .., this represents a ..% fall. Given the above difference-in-difference result, we believe that a significant reduction analyst coverage could have contributed to the reduced information production, which in turn could have influenced valuation of the affected stocks.

G.3 Price Efficiency- Earnings Surprise

We now test of derivative delisting leads to increase in earnings surprise. We estimate regression equation 1 with SUE of a quarter as the dependent variable. The results are reported in Table 9.

The table shows the impact of derivative exclusion on price efficiency of a stock. Price

Table 9: Analyst Following Before And After Exclusion

The table shows the impact of derivative exclusion on analyst coverage of a stock. The total sample includes firms that are excluded from the derivative segment and those that continue to have derivatives traded on them even after the SEBI order. The dependent variable is the number of analysts following an firm i as at end of quarter j when compared to the same as at the end of quarter j-1. The independent variable after is a dummy variable that takes the value of 1 if quarter i lies in the post delisting period and zero otherwise. Treat is a dummy that takes the value of 1 if the stock i under consideration is an excluded stock and zero otherwise. The main independent variable of interest is the interaction between After and Treat dummies. We also include firm level controls for size(market cap), insider holding(MWPL) and liquidity (MQSOS). In column 1 and 2 we do no include any fixed effects. In column 3, we include firm fixed effects and in column 4, we include both firm and time fixed effects. T-statistics are reported in parenthesis. ***, **, * represents statistical significance at the 1%, 5% and 10% levels.

	(1)	(2)	(3)	(4)
VARIABLES	Med_reco	Med_reco	Med_reco	Med_reco
treatment	-18.272***	-10.744***		
	(1.861)	(1.852)		
post_event	3.354***	3.847***	3.872***	7.680***
	(0.536)	(0.454)	(0.459)	(0.665)
int_treat_post	-4.343***	-2.551***	-2.755***	-2.265**
	(1.080)	(0.967)	(0.985)	(0.963)
Observations	3,178	3,125	3,125	3,125
R-squared			0.219	0.364
Number of observations	206	204	204	204
Adj R-squared	0.0990	0.216	0.219	0.364
Control Vars		YES	YES	YES
Time FE	NO	NO	NO	YES
Firm FE	NO	NO	YES	YES

Table 10: Earnings Suprise before and After the Event

The table shows the impact of derivative exclusion on price efficiency of a stock. Price efficiency is measured using Standardised Unexpected Earnings method as in (**?**. The total sample includes firms that are excluded from the derivative segment and those that continue to have derivatives traded on them even after the SEBI order. The dependent variable is SUE of a firm i in a quarter j. The independent variable after is a dummy variable that takes the value of 1 if quarter i lies in the post delisting period and zero otherwise. Treat is a dummy that takes the value of 1 if the stock i under consideration is an excluded stock and zero otherwise. The main independent variable of interest is the interaction between After and Treat dummies. We also include firm level controls for size(market cap), insider holding(MWPL) and liquidity (MQSOS). In column 1 we do no include any fixed effects. In column 2, we include firm fixed effects and in column 3, we include both firm and time fixed effects. T-statistics are reported in parenthesis. ***, **, * represents statistical significance at the 1%, 5% and 10% levels.

VARIABLES	sue_pbdit	sue_pbdit	sue_pbdit	sue_pbdit
Treatment	-0.051	-0.299		
	(0.170)	(0.212)		
Post_event	-0.143	-0.191	-0.194	-0.294
	(0.131)	(0.125)	(0.130)	(0.200)
Int_treat_post	1.347*	1.373*	1.132*	1.158*
	(0.775)	(0.827)	(0.651)	(0.662)
Observations	3,272	3,191	3,191	3,191
R-squared			0.005	0.010
Number of company_id	206	204	204	204
Adj R-squared	0.00287	0.00354	0.00489	0.00952
Control Vars		YES	YES	YES
Time FE				YES
Firm FE			YES	YES

efficiency is measured using Standardised Unexpected Earnings method as in (?. The total sample includes firms that are excluded from the derivative segment and those that continue to have derivatives traded on them even after the SEBI order. The dependent variable is SUE of a firm i in a quarter j. The independent variable after is a dummy variable that takes the value of 1 if quarter i lies in the post delisting period and zero otherwise. Treat is a dummy that takes the value of 1 if the stock i under consideration is an excluded stock and zero otherwise. The main independent variable of interest is the interaction between After and Treat dummies. We also include firm level controls for size(market cap), insider holding(MWPL) and liquidity (MQSOS). In column 1 we do no include any fixed effects. In column 2, we include firm fixed effects and in column 3, we include both firm and time fixed effects.

Our focus is the interaction between post delisting period dummy and the treatment dummy. In all specification, the interaction term is positive and statistically significant. In other words, earnings surprise increases post derivative delisting. This indicates that significant price efficiency is lost post delisting. The above result supports our hypothesis that derivatives contribute towards enhancing price efficiency.

H Change in Liquidity

As we have discussed in Section V, a second channel that could influence valuation is the change in liquidity. Given the short selling constraints in the spot market, it is difficult to argue that all traders who were trading in derivatives seamlessly move to spot markets post exclusion of the stock from the derivative segment. As per procedure described in Section V, we test if the trading volume in spot markets change delisting. We report the results of estimating equation (6) in table 9 below.

It is important to note that actual derivative trading continued for 45 days after the event date as existing contracts were kept active even after notification. Keeping this in mind, we devise our event windows. Each column in the table represents a event window. Column 2 covers an interval comprising of the event day and 3 days immediately after the event. We chose this window because the July contract was due for expiry on 25th of

Table 11: Market Reaction

This is a cross sectional regression with abnormal volume of a stock i during a window j being the dependent variable. The total sample includes retained as well as excluded stocks. Treat is a dummy that takes the value of one for excluded stocks and zero for other stocks. We include firm level control variables similar to the ones used before. We report different intervals (specified in top row) of time around the event day in different columns.

Interval	_-10_-1	_0_3	4_27	28_46	47_64	1_120
VARIABLES	turnover	turnover	turnover	turnover	turnover	turnover
Treatment	-0.035	0.338*	-0.038	-0.041	-0.548***	-0.291***
	(0.078)	(0.192)	(0.099)	(0.218)	(0.174)	(0.104)
Firm Level Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	200	200	200	200	200	200
R-squared	0.020	0.069	0.016	0.029	0.096	0.056
Adj R-squared	-0.000111	0.0496	-0.00420	0.00916	0.0778	0.0370

July (3days from the event date of 22nd July). We expect a spike in volume immediately after announcement due to incremental trades aimed at closing existing open positions. We do find an immediate spike of 33.8% (column 2) in volume from the event date to the date of expiry. Please note that the next two months contract were kept open as they existed before the date of SEBI announcement. Therefore, we report, in columns 3 and 4, abnormal volumes for the excluded stocks for next two months separately. We do not find any statistically significant jump in volume in the next two months. Finally, in column 5, we look at volume reaction after 46th day, when all derivative contracts are eliminated. Here we find that during one month period following the 46th day, volumes in the spot market fall by a whopping 54.8%. In other words, volumes in the spot market fall by more than a half post derivative exclusion.

It is possible to argue that some of the fall is merely a reversal of the 33.8% increase (reported in column 2) seen immediately after the announcement by SEBI. In order to asses the overall impact of derivative exclusion on trading volume in the spot market, we report the abnormal volumes for the excluded stocks when compared to retained stocks for a period of 120 days from the date of exclusion. Here, again we find that volumes decline by 28.3% for the excluded stocks when compared to retained stocks. The number is comparable to the estimate made by Berkman, Eleswarapu, et al. (1998). Above results show that derivatives indeed improve liquidity in the spot market and hence add value to a stock.

VII Conclusion

In this paper we revisit the question of whether derivatives are passive securities. Most studies in this area use derivatives listing or delisting as events to study the impact of derivatives on stock characteristics. However, with Mayhew and Mihov (2004) and Danielsen, Van Ness, and Warr (2007) showing that the decision of an exchange to list derivatives could be influenced by expected trends in fundamentals, a question mark has risen over the findings of the extant literature which has used derivative listing as an event to study the impact of derivatives. Some studies have tried to overcome the same by considering stocks with high derivative trading as the ones who are likely to have maximum impact of derivatives when compared to stock with low derivative trading. However, decision to trade derivatives on stock might be influenced a number of endogenous factors. And finally and most importantly, there is no consensus regarding the possible impact of derivatives on stock characteristics ; some studies show positive impact Roll, Schwartz, and Subrahmanyam (2009); Conrad (1989), while some others show negative impact (Danielsen and Sorescu (2001) and some other showing no impact (Vijh (1990)).

We contribute to the literature by examining the impact of derivatives on stock fundamentals using a exogenous event. On July 22nd 2012, India's market regulator issued an order tightening the eligibility criteria for continuance in the derivative segment. These criteria were based on amount of outside ownership (MWPL) and trading volume (MQSOS) during the preceeding six months. The applicable limits with respect to both criteria were raised substantially. Thus the change in criteria was based on purely historical information.

Using the above natural experiment, we find that derivatives indeed add value. Derivative exclusion leads to 4.82% negative cumulative abnormal returns for the excluded stocks in the 5 days after exclusion. Average Tobin's Q is lower by 6.3% in post exclusion period when compared to pre exclusion period. We then investigate the channels of such influence. We show that derivatives impact valuation by enhancing price efficacy as well as liquidity. Derivative exclusion leads lessor reliance on the part of corporate managers on stock returns in their capital expenditure decisions and also lower analyst following. Trading volume in the spot market also goes down for the affected stocks.

We rule out regulatory targeting by performing a number of placebo tests. First, we show that there is no trend in the attributes used as benchmarks for exclusion. Especially there is no specials jumps for excluded stocks towards the end of the estimation period. Second, we perform several placebo tests to rule out alternative explanations. Third, we estimate our main result using only those stocks that get excluded because of a relatively sticky criteria of insider ownership. Finally, we use synthetic control method to create a valid counter factual.

References

Abadie, A., A. Diamond, and J. Hainmueller (2010): "Synthetic control methods for comparative case studies: Estimating the effect of Californias tobacco control program," *Journal of the American Statistical Association*, 105(490).

Abadie, A., and J. Gardeazabal (2003): "The economic costs of conflict: A case study of the Basque Country," *American economic review*, pp. 113–132.

Acemoglu, D., S. Johnson, A. Kermani, J. Kwak, and T. Mitton (2013): "The value of connections in turbulent times: Evidence from the United States," Discussion paper, National Bureau of Economic Research.

Admati, A. R., and P. Pfleiderer (1988): "A theory of intraday patterns: Volume and price variability," *Review of Financial studies*, 1(1), 3–40.

Amihud, Y., and H. Mendelson (1986): "Asset pricing and the bid-ask spread," *Journal of financial Economics*, 17(2), 223–249.

Asquith, P., M. B. Mikhail, and A. S. Au (2005): "Information content of equity analyst reports," *Journal of financial economics*, 75(2), 245–282.

Berkman, H., V. R. Eleswarapu, et al. (1998): "Short-term traders and liquidity: a test using Bombay Stock Exchange data," *Journal of Financial Economics*, 47(3), 339–355.

Bertrand, M., E. Duflo, and S. Mullainathan (2002): "How much should we trust differences-in-differences estimates?," Discussion paper, National Bureau of Economic Research.

Bhue, G. S., N. Prabhala, and P. Tantri (2015): "Creditor Rights and Relationship Banking: Evidence from a Policy Experiment," .

Black, F., and M. Scholes (1973): "The pricing of options and corporate liabilities," *The journal of political economy*, pp. 637–654.

Blanco, I., and D. Wehrheim (2015): "The Bright Side of Financial Derivatives: Options Trading and Innovation," *Available at SSRN 2586441*.

Bloomfield, R., M. Ohara, and G. Saar (2009): "How noise trading affects markets: An experimental analysis," *Review of Financial Studies*, 22(6), 2275–2302.

Boehmer, E., J. Masumeci, and A. B. Poulsen (1991): "Event-study methodology under conditions of event-induced variance," *Journal of Financial Economics*, 30(2), 253–272.

Bose, S. (2007): "Commodity Futures Market in India: A Study of Trends in the Notional Multi-Commodity Indices," *Money & Finance, ICRA Bulletin*, 3(3).

Cao, H. H. (1999): "The effect of derivative assets on information acquisition and price behavior in a rational expectations equilibrium," *Review of Financial Studies*, 12(1), 131–163.

Chakravarty, S., H. Gulen, and S. Mayhew (2004): "Informed trading in stock and option markets," *The Journal of Finance*, 59(3), 1235–1258.

Chen, Q., I. Goldstein, and W. Jiang (2007): "Price informativeness and investment sensitivity to stock price," *Review of Financial Studies*, 20(3), 619–650.

Conrad, J. (1989): "The price effect of option introduction," *The Journal of Finance*, 44(2), 487–498.

Damodaran, A., and J. Lim (1991): "The effects of option listing on the underlying stocks' return processes," *Journal of Banking & Finance*, 15(3), 647–664.

Danielsen, B. R., and S. M. Sorescu (2001): "Why do option introductions depress stock prices? A study of diminishing short sale constraints," *Journal of Financial and Quantitative Analysis*, 36(04), 451–484.

Danielsen, B. R., B. F. Van Ness, and R. S. Warr (2007): "Reassessing the impact of option introductions on market quality: A less restrictive test for event-date effects," *Journal of Financial and Quantitative Analysis*, 42(04), 1041–1062.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990): "Noise trader risk in financial markets," *Journal of political Economy*, pp. 703–738.

Diamond, D. W., and R. E. Verrecchia (1987): "Constraints on short-selling and asset price adjustment to private information," *Journal of Financial Economics*, 18(2), 277–311.

Durnev, A., R. Morck, and B. Yeung (2004): "Value-enhancing capital budgeting and firm-specific stock return variation," *The Journal of Finance*, 59(1), 65–105.

Easley, D., M. O'hara, and P. S. Srinivas (1998): "Option volume and stock prices: Evidence on where informed traders trade," *The Journal of Finance*, 53(2), 431–465.

Gopalan, R., V. Nanda, and A. Seru (2007): "Affiliated firms and financial support: Evidence from Indian business groups," *Journal of Financial Economics*, 86(3), 759–795.

Harris, L., and E. Gurel (1986): "Price and volume effects associated with changes in the S&P 500 list: New evidence for the existence of price pressures," *The Journal of Finance*, 41(4), 815–829.

Ho, L.-C. J., and C.-S. Liu (1997): "A reexamination of price behavior surrounding option introduction," *Quarterly Journal of Business and Economics*, pp. 39–50.

Hu, J. (2014): "Does option trading convey stock price information?," *Journal of Financial Economics*, 111(3), 625–645.

Jain, A., P. L. Tantri, and R. S. Thirumalai (2014): "Do Demand Curves for Stocks Slope Down?: Evidence from a Regulation Induced Supply Shock," *Evidence from a Regulation Induced Supply Shock (November 1, 2014)*.

Karlan, D., and J. Zinman (2009): "Expanding credit access: Using randomized supply decisions to estimate the impacts," *Review of Financial studies*, p. hhp092.

Kaserer, C., and B. Moldenhauer (2008): "Insider ownership and corporate performance: evidence from Germany," *Review of Managerial Science*, 2(1), 1–35.

Kaul, A., V. Mehrotra, and R. Morck (2000): "Demand curves for stocks do slope down: New evidence from an index weights adjustment," *The Journal of Finance*, 55(2), 893–912.

Khanna, T., and K. Palepu (2000): "Is group affiliation profitable in emerging markets? An analysis of diversified Indian business groups," *The Journal of Finance*, 55(2), 867–891.

Kolari, J. W., and S. Pynnönen (2010): "Event study testing with cross-sectional correlation of abnormal returns," *Review of Financial Studies*, p. hhq072.

Kumar, R., A. Sarin, and K. Shastri (1998): "The impact of options trading on the market quality of the underlying security: An empirical analysis," *The Journal of Finance*, 53(2), 717–732.

LONG, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990): "Positive feedback investment strategies and destabilizing rational speculation," *the Journal of Finance*, 45(2), 379–395.

Mayhew, S., and V. Mihov (2004): "How do exchanges select stocks for option listing?," *The Journal of Finance*, 59(1), 447–471.

Mayhew, S., and V. T. Mihov (2005): "Short sale constraints, overvaluation, and the introduction of options," in *AFA 2005 Philadelphia Meetings*.

Mehta, P., S. Mullainathan, and M. Bertrand (2002): "FERRETING OUT TUNNELING: AN APPLICATION TO INDIAN BUSINESS GROUPS," *Quaterly journal of economics*, (1), 121–148.

Muravyev, D., N. D. Pearson, and J. P. Broussard (2013): "Is there price discovery in equity options?," *Journal of Financial Economics*, 107(2), 259–283.

Naiker, V., F. Navissi, and C. Truong (2012): "Options trading and the cost of equity capital," *The Accounting Review*, 88(1), 261–295.

Ni, S. X., N. D. Pearson, and A. M. Poteshman (2005): "Stock price clustering on option expiration dates," *Journal of Financial Economics*, 78(1), 49–87.

Pearson, N. D., A. M. Poteshman, and J. S. White (2007): "Does Option Trading Have a Pervasive Impact on Underlying Stock Prices?," in *AFA 2008 New Orleans Meetings Paper*.

Roll, R., E. Schwartz, and A. Subrahmanyam (2009): "Options trading activity and firm valuation," *Journal of Financial Economics*, 94(3), 345–360.

——— (2010): "O/S: The relative trading activity in options and stock," *Journal of Financial Economics*, 96(1), 1–17.

Ross, S. A. (1976): "Options and efficiency," *The Quarterly Journal of Economics*, pp. 75–89.

Sahoo, P., and R. Kumar (2009): "Efficiency and futures trading-price nexus in Indian commodity futures markets," *Global Business Review*, 10(2), 187–201.

Shleifer, A., and R. W. Vishny (1997): "The limits of arbitrage," *The Journal of Finance*, 52(1), 35–55.

Vig, V. (2013): "Access to collateral and corporate debt structure: Evidence from a natural experiment," *The Journal of Finance*, 68(3), 881–928.

Vijh, A. M. (1990): "Liquidity of the CBOE equity options," *The Journal of Finance*, 45(4), 1157–1179.